



Dayananda Sagar College of Engineering
An Autonomous Institute Affiliated to VTU, Belagavi
Department of Artificial Intelligence and Machine Learning

DISCRETE MATHEMATICAL STRUCTURES

Course Code	22MAT41A	Credits:	03
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	BSC	Exam Hours	03

Course Objectives	
1	To introduce the basics of Set Theory, Number Theory and some of its applications.
2	To understand and apply problem-solving skills by using truth tables, basic logical connectives and proofs.
3	To understand the Properties of Relations and various types of functions also graph theory concepts to apply some of its applications in computer science.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Understand basic concepts of Set Theory & Number Theory, Fundamental logics, functions, Relations and Graph Theory.	L2
CO2	Analyze the logic for validity of the given arguments, sets and perform operations and algebra on sets, properties of Functions, Relations and Graph Theory.	L4
CO3	Apply the concepts of Relations, Hasse Diagram and Poset, Mathematical Logics, Functions and Number Theory, Graph theory in Engineering Problems.	L3
CO4	Evaluate functions like Stirling numbers using fundamental principles, logical propositions via truth tables, combinations and permutations on sets, relations and digraphs, trees in Engineering field.	L3
CO5	Solve the engineering problems using mathematical logic, recurrence relations, even and odd integers, rational and irrational numbers and divisibility using Number Theory, connection between bijective functions and real life problems using graphs and trees.	L4

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	-	3	-	-	-	-	-	-	-	-	-	-	-	-	-

Module	Topic	Hours	Assessment	CO
1	Set Theory and Number Theory: Sets and Subsets, Set Operations and the Laws of Set Theory, Addition Principles, Concept of Number Theory-Simple ProblemsCase study. Application: Representation of Set Theory problems using Venn diagrams Pedagogy: Chalk and talk/PowerPoint Presentation/Videos	8	CIE+SEE+QUIZ	CO1-CO5 L1-L4



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	Web Link: https://onlinecourses.nptel.ac.in/noc20_ma42/preview Video Lectures :			
2	Fundamentals of Logic: Basic Connectives and Truth Tables, Logical Equivalence – The Laws of Logic, Logical Implication– Rules of Inference, Quantifiers with one variable- Case study. Application: AND- OR gate problems for real time. Pedagogy: Chalk and talk/PowerPoint Presentation/Videos Web Link: https://nptel.ac.in/courses/106102013 Video Lectures :	8	CIE+SEE+QUIZ	CO1- CO5 L1-L4
3	Relations: Properties of Relations, Computer Recognition: ZeroOne Matrices and Directed Graphs, Partial Orders – Hasse Diagrams, Equivalence Relations and Composition of RelationsCase study. Application: Recurrence relation to solve Tower of Hanoi, Fibonacci numbers and Rabbit Population Problem. Pedagogy: Chalk and talk/PowerPointPresentation/Videos Web Link: https://archive.nptel.ac.in/courses/111/107/111107058/ Video Lectures:	8	CIE+SEE+QUIZ	CO1- CO5 L1-L4
4	Functions: Cartesian Products, Functions – Plain and One-to- One, Onto Functions – Stirling Numbers of the Second Kind, Function Composition- Case study. Application: Pigeon Hole Principle for allocations. Pedagogy: Chalk and talk/PowerPoint Presentation/Videos Web Link: https://nptel.ac.in/courses/111107058 Video Lectures :	8	CIE+SEE+QUIZ	CO1- CO5 L1-L4
5	Graph Theory & Its Applications: Definitions and Examples, Sub graphs, Complements, and Graph Isomorphism, Euler Trails and Circuits, Hamiltonian paths, cycles, Basics of Graph Coloring-Case study. Application: Coloring of Maps, Solving Sudoku, Coloring of graphs resulting in Bipartite graphs. Pedagogy: Chalk and talk/PowerPointPresentation/Videos Web Link: https://onlinecourses.nptel.ac.in/noc20_ma05/preview Video Lectures:	8	CIE+SEE+QUIZ	CO1- CO5 L1-L4
	Student development program: Workshop on Lattice theory. Self-Study component: Module 1: Cardinality Properties of Set Theory and Applications of Number Theory Module 2: Validity of arguments in propositional calculus and Uses of Logics Module 3: Classification of function and Inverse Function. Module 4: Relational matrix and graph of a relation Module 5: Graph Homomorphism and its applications.			

Text Books	
1	Ralph P. Grimaldi: Discrete and Combinatorial Mathematics, 5th Edition, PearsonEducation. 2006, ISBN-10. 9788177584240.
2	B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 43rd Edition, 2014 June, ISBN: 9788174091956.
3	David Burton, Elementary Number Theory 7th Edition, Mc Graw Hill Higher Education 2006 ISBN 978-0-07-338314-9.



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OPERATING SYSTEMS

Course Code	22AI42	Credits:	03
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	03

Course Objectives	
1	To understand the importance of operating systems in utilizing the hardware and software resources along with the protection and security mechanisms.
2	To comprehend and analyze various resource management techniques.
3	To implement resource management algorithms.
4	To gain knowledge on the various protection and security mechanisms provided by operating system.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Gain extensive knowledge on principles and modules of Operating Systems	L3
CO2	Apply suitable algorithms for Process Scheduling and Process Synchronization	L3
CO3	Apply suitable techniques for efficient Memory management and Storage management	L3
CO4	Realize the different concepts of OS in platform of usage through case studies.	L4

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	3
CO4	-	3	-	-	-	-	-	-	-	-	-	-	2	-	-

Module	Topic	Hours	Assessment	CO
1	Overview-Introduction: What operating systems do; Operating System Structure, Operating System operations. System Structures: Operating System Services; System calls; Types of System calls.	8	CIE+SEE+QUIZ	CO1
2	Process Management-Process Concept: Process concept; Process scheduling; Process Scheduling: Basic concepts, process creation and termination; scheduling criteria; Scheduling algorithms: FCFS, Pre-emptive and Non-preemptive SJF, Priority Scheduling and Round Robin Scheduling	8	CIE+SEE+QUIZ	CO2



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3	Process Coordination- Synchronization: The Critical section problem, Peterson solution, mutex locks, Semaphores; Classical problems of synchronization: Bounded-Buffer Problem, Reader's-Writer's Problem. Deadlocks: System model; Deadlock characterization; Methods for handling deadlocks; Deadlock prevention; Deadlock avoidance; Deadlock detection and recovery from deadlock.	8	CIE+SEE+QUIZ	CO2
4	Memory Management-Memory Management Strategies: Background; Contiguous memory allocation; Paging; Virtual Memory management: Background; Demand paging; Page replacement Algorithms: FIFO, LRU, OPTIMAL	8	CIE+SEE+QUIZ	CO3
5	Storage Management-File System: File concept; Access methods; Protection. Secondary Storage Structures: Overview of Mass storage structures; Disk scheduling algorithms. System Protection: Goals of protection, Principles of protection, Domain of protection, Access Matrix.	8	CIE+SEE+QUIZ	CO3

Text Books	
1	Abraham Silberschatz, Peter Baer Galvin, Greg Gagne: Operating System Principles, 8th edition, Wiley India, 2009

Reference Books	
1	D.M. Dhamdhare: Operating systems - A concept based Approach, 2nd Edition, Tata McGraw- Hill, 2002
2	P.C.P. Bhatt: Introduction to Operating Systems: Concepts and Practice, 2nd Edition, PHI, 2008. Harvey M Deital: Operating systems, 3rd Edition, Pearson Education, 1990.
3	Harvey M Deital: Operating systems, 3rd Edition, Pearson Education, 1990



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DESIGN & ANALYSIS OF ALGORITHMS

Course Code	22AI43	Credits:	04
L:T:P:S	3:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	IPCC	Exam Hours	03

Course Objectives	
1	To learn the methods for analyzing algorithms and evaluating their performance.
2	To demonstrate the efficiency of algorithms using asymptotic notations.
3	To solve problems using various algorithm design methods, including brute force, greedy, divide and conquer, decrease and conquer, transform and conquer, dynamic programming, backtracking, and branch and bound.
4	To learn the concepts of P and NP complexity classes.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Apply Brute Force Techniques to solve the Numerical, String and Graph problems.	L3
CO2	Apply Recursive Algorithms towards optimal solution for real time problems.	L3
CO3	Analyze the time complexity of the P Class problems.	L4
CO4	Analyze the time complexity of the NP and NP Hard Class problems	L4
CO5	Implement and analyze the efficiency of algorithmic design techniques in solving Real time problems using open source tools.	L4
CO6	Write clear and concise experiment reports that detail the methods, results, and Analysis of algorithm techniques.	L4

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO2	3		-	-	-	-	-	-	-	-	-	-	2	-	-
CO3	-	3	-	-	-	-	-	-	-	-	-	-	-	-	3
CO4	-	3	-	-	-	-	-	-	-	-	-	-	2	-	-
CO5	-	3	-	-	-	-	-	-	-	-	-	-	3	-	-
CO6	-	3	-	-	-	-	-	-	2	2	-	-	-	-	2

Module	Topic	Hours	Assessment	CO
1	Introduction: What is an Algorithm?, Algorithm Specification , Analysis Framework), Performance Analysis: Space complexity, Time complexity . Asymptotic Notations: Big-Oh notation (O), Omega notation (Ω), Theta notation (Θ), and Little-oh notation (o), Mathematical analysis of NonRecursive and recursive Algorithms with Examples . Brute Force Approaches: Selection Sort and Bubble Sort, Sequential Search and Brute Force String Matching, Exhaustive Search (Travelling Salesman problem and Knapsack Problem).	8	CIE+SEE+QUIZ	CO1- CO4



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2	Divide and Conquer: General method, Binary search, Recurrence equation for divide and conquer, Finding the maximum and minimum , Merge sort, Quick sort , Strassen's matrix multiplication , Advantages and Disadvantages of divide and conquer. Decrease and Conquer Approach: Insertion Sort, , Topological Sort	8	CIE+SEE+QUIZ	CO1-CO4
3	Greedy Method: General method, Coin Change Problem, Knapsack Problem, Job sequencing with deadlines. Minimum cost spanning trees: Prim's Algorithm, Kruskal's Algorithm. Single source shortest paths: Dijkstra's Algorithm. Optimal Tree problem: Huffman Trees and Codes Transform and Conquer Approach: Heaps and Heap	8	CIE+SEE+QUIZ	CO1-CO4
4	Dynamic Programming: General method with Examples, Transitive Closure: Warshall's Algorithm, All Pairs Shortest Paths: Floyd's Algorithm, Optimal Binary Search Trees, Knapsack problem, Bellman-Ford Algorithm, Travelling Sales Person problem.	8	CIE+SEE+QUIZ	CO1-CO4
5	Backtracking: General method, N-Queens problem, Sum of subsets problem, Graph coloring, Hamiltonian cycles. Branch and Bound: Assignment Problem, Travelling Sales Person problem, 0/1 Knapsack problem: LC Branch and Bound solution, FIFO Branch and Bound solution. NP Complete and NP-Hard problems: Basic concepts, nondeterministic algorithms, P, NP, NP-Complete, and NP-Hard classes.	8	CIE+SEE+QUIZ	CO1-CO4

Text Books	
1	Introduction to the Design and Analysis of Algorithms, By Anany Levitin, 3rd Edition (Indian), 2017, Pearson.
2	Computer Algorithms, Ellis Horowitz, Satraj Sahni and Rajasekaran, 2nd Edition, 2014, Universities Press.

Reference Books	
1	Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronal L. Rivest, Clifford Stein, 3rd Edition, PHI.
2	Design and Analysis of Algorithms, S. Sridhar, Oxford (Higher Education)



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MACHINE LEARNING ESSENTIALS

Course Code	22AI44	Credits:	04
L:T:P:S	3:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	IPCC	Exam Hours	03

Course Objectives	
1	Understand the basics of machine learning, including different types of learning and how models work.
2	Practice using important machine learning algorithms like linear regression and decision trees with real-world data.
3	Develop skills to clean and prepare data for analysis, including techniques for handling missing values and creating useful features.
4	Learn how to assess model performance and make improvements to increase accuracy and effectiveness.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Understand the foundation of Machine learning and statistical methods in Machine Learning	L3
CO2	Apply linear and logistic regression techniques in appropriate problem scenarios	L3
CO3	Use decision trees to solve real world problems	L4
CO4	Analyze the dimensionalities of various classification and clustering techniques	L4
CO5	Apply data analytics and feature engineering techniques on dataset	L4
CO6	Apply regression analysis, classification, clustering, time series analysis and ensemble learning techniques	L4

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3		-	-	-	-	-	-	-	-	-	-	2	-	-
CO3	3	-	-		-	-	-	-	-	-	-	-	-	3	3
CO4	-	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	3	-	-	-	-	-	-	-	2	2	-	-	-	2	-

Module	Topic	Hours	Assessment	CO
1	Introduction: Applications of machine learning, Branches of ML, Types of Machine Learning, Machine Learning Process, Data types and Data Analytics for Machine learning, Statistical tests: Mean, Median, Mode, Standard Deviation, 5-point Summary, Student t test, Z test, Chi- square test, Correlation, ANOVA, Hypothesis Testing	8	CIE+SEE+QUIZ	CO1



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2	Linear Regression: Introduction, Advantages and Disadvantages, Applications, Simple Linear Regression, Train Test split, Cost function, Multiple Linear Regression, Model Interpretation, Goodness of fit, Model Selection, Assumptions of Linear Regression, Feature Engineering - Categories as predictors – Dummy Variable, Label Encoding, Interaction, Scaling, Standardisation, Missing value treatment, Outlier Treatment, Feature Extraction, Transformation	8	CIE+SEE+QUIZ	CO2
3	Logistic regression: Why not linear regression? (S Shaped Curve) Need for logistic regression, Introduction, Advantages and Disadvantages, Applications, The Logistic model: Estimating the Regression Coefficients, Making Predictions. Goodness of fit-i) Confusion matrix ii) Precision iii) Recall iv) F1 Score v) ROC / AUC Handling Imbalance class	8	CIE+SEE+QUIZ	CO2
4	Decision Tree: Decision Tree learning Model, Entropy, GINI, Working of single Decision Tree, CART and CHAID algorithm, validating and Pruning of Decision Trees, Probability based Learning, Bayes Theorem, Classification using Bayes Model, Naïve Bayes Algorithm for Continuous Attributes, Other Naïve Bayes Classifiers	8	CIE+SEE+QUIZ	CO3
5	Clustering: Introduction to clustering, Applications of clustering, Distance Measures, Euclidean, Hamming, Manhattan, Minkowski. Hierarchical Clustering, K Means Algorithm, K Mode, K Prototype, DBSCAN (Density based Clustering), Cluster validation Dimensionality Reduction - PCA (Principal Component Analysis)	8	CIE+SEE+QUIZ	CO4

Text Books	
1	Mitchell, T.M. and Tom, M. (1997) Machine Learning. McGraw-Hill, New York.



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SOFTWARE ENGINEERING

Course Code	22AI461	Credits:	03
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	ESC	Exam Hours	03

Course Objectives	
1	Outline software engineering principles and activities involved in building large software programs.
2	Introduce Software design using UML tools and various system models.
3	Introduce Agile Software development and Agile methods.
4	Introduce Software Testing and Software Evolution processes in detail.
5	Explain project planning, Cost estimation techniques and Software quality, standards and metrics.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Illustrate the fundamentals of Software Engineering, Software process models, and Requirements Engineering.	L2
CO2	Apply Agile software development and Agile methods.	L3
CO3	Apply and Analyse Software Testing methods.	L3
CO4	Discuss the Project Planning process, Software Quality standards, and business strategies.	L3

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	3	-
CO4	3	-	-	-	-	-	-	-	-	-	-	-	-	-	2

Module	Topic	Hours	Assessment	CO
1	Introduction: Professional Software Development - Software Engineering, Case studies. Software Processes: Software process models - Waterfall Model, Incremental Model and Spiral Model, Software Process Activities Requirement Engineering; Requirements Specification, Requirement elicitation and analysis, Requirements Validation, Requirements Management, Functional Specification and Non-Functional Requirements.	8	CIE+SEE+QUIZ	CO1
2	Software Design and Modeling Concepts: What is OO orientation? What is OO development, OO themes, Modeling as design technique, modeling, abstraction, The three models, object and class concepts , Links and association concepts, A sample class model. System Models: Context models. Interaction models. Structural models. Behavioral models. Model-driven engineering	8	CIE+SEE+QUIZ	CO2



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3	Agile Software Development: Agile methods, Plan-driven and agile development, Extreme programming, Agile project management, Scaling agile methods. SCRUM Methodology, SCRUM. Project Scheduling: Basic Concepts, Project Scheduling, Defining a Task set for the software Project, Defining a Task network, Scheduling.	8	CIE+SEE+QUIZ	CO2,3
4	Software Testing: Development testing, Test-driven development, Release testing, User testing, Dataflow testing and Path Testing, Case Study: Commission Problem Software Evolution: Evolution processes, Program evolution dynamics, Software maintenance, Legacy system management	8	CIE+SEE+QUIZ	CO1,4
5	Project planning: Software pricing, Plan-driven development, Project scheduling, Agile planning, Estimation techniques, COCOMO cost modeling. Case Study on Project Planning and Management Quality management: Software quality, Software standards, Reviews and inspections, Software measurement and metrics Business Strategy: The concept of strategy, Introduction and objectives, Role of Strategy in success, Framework of strategy analysis, Brief history of Business strategy, The Strategy process.	8	CIE+SEE+QUIZ	CO4

Text Books

1	“Software Engineering”, International Computer Science Series, Ian Somerville, Pearson Education, 10th Edition, 2020.
2	Michael Blaha, James Rumbaugh, Object Oriented Modelling and Design with UML, , Pearson Education, 2ndEdition, 2017.
3	“Software Testing – Principles and Practices”, Srinivasan Desikan and Gopalaswamy Ramesh, Pearson, 2018.

Reference Books

1	“Software Engineering: Principles and Practice”, Hans van Vliet, Wiley India, 3rd Edition, 2018.
2	“Effective DevOps: Building a Culture of Collaboration, Affinity, and Tooling” by By Jennifer Davis, Ryn Daniels, O’Reilly Publications, 2018
3	“Software Engineering: A Practitioner’s Approach”, Roger S Pressman, McGraw Hill, 6th Edition 2015.
4	The SCRUM Primer, Ver2.0, http://www.goodagile.com/scrumprimer/scrumprimer20.pdf .



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DATAVISUALIZATION LABORATORY

Course Code	22AIL45	Credits:	01
L:T:P:S	0:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	PCCL	Exam Hours	03

Course Objectives	
1	To Understand the importance of data visualization for business intelligence and decision making.
2	Know approaches to understand visual perception.
3	Learn about categories of visualization and application areas.
4	Familiarize with the data visualization tools.
5	Gain knowledge of effective data visuals to solve workplace problems.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Apply data visualization techniques to diverse datasets to create appropriate visualizations and to identify both positive and negative impacts of data-informed decision.	L3
CO2	Analyze the data visualization tools to represent and interpret the results of stories created using various features.	L4
CO3	Design interactive visualization by analysing the data visualization tools and libraries that allow users to explore and gain insights from data.	L4
CO4	Write clear and concise report to document experimental procedures, observations, and results accurately, adhering to established standards.	L5

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	-	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO4	-	-	-	3	-	-	-	-	-	-	-	-	-	-	2

Module	Topic	Hours	Assessment	CO
1	TFL Bus Safety: i. Create a bar chart on boroughs field to visualize the trend in the count. ii. Create a line chart for date of incidence for each month in a quarter, comment on possibilities and suitability of different charts for this timeline. iii. In above question, apply formatting to display the first letter of the month on X-axis. iv. Create tree maps of all the data fields except date & year and comment on significance of tree map. v. Create an interactive dashboard for the above data.	2	CIE+SEE+QUIZ	CO1,2,3,4



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2	GDP dataset: - Visualize the countries data given in the dataset with respect to latitude and longitude along with country name using symbol maps. - Create a bar graph to compare GDP of Belgium between 2006 – 2026. - Using pie chart, visualize the GDP of India, Nepal, Romania, South Asia, Singapore by the year 2010. - Visualize the countries Bhutan & Costa Rica competing in terms of GDP. - Create a scatter plot or circle views of GDP of Mexico, Algeria, Fiji, Estonia from 2004 to 2006. - Build an interactive dashboard.	2	CIE+SEE+QUIZ	CO1,2 ,3,4
3	Customer Analysis dataset: - Create a choropleth map (fill the map) to spot the special trends to show the state which has the highest revenue. - Create a line chart to show the revenue based on the month of the year. - Create a bin of size 10 for the age measure to create a new dimension to show the revenue. - Create a donut chart view to show the percentage of revenue per region by creating zero access in the calculated field. - Create a butterfly chart by reversing the bar chart to compare female & male revenue based on product category. - Create a calculated field to show the average revenue per state & display profitable & non-profitable state. - Build a dashboard.	2	CIE+SEE+QUIZ	CO1,2 ,3,4
4	Amazon Prime Dataset: - Create a Donut chart to show the percentage of movie and tv shows - Create a area chart to shows by release year and type. - Create a horizontal bar chart to show Top 10 genre. - Create a map to display total shows by country. - Create a text sheet to show the description of any movie/movies. - Build an interactive Dashboard.	2	CIE+SEE+QUIZ	CO1,2 ,3,4
5	HR Dataset: - Create KPI to show employee count, attrition count, attrition rate, attrition count, active employees, and average age. - Create a Lollipop Chart to show the attrition rate based on gender category. - Create a pie chart to show the attrition percentage based on Department Category - Create a bar chart to display the number of employees by Age group, - Create a highlight table to show the Job Satisfaction Rating for each job role based on employee count. - Create a horizontal bar chart to show the attrition count for each Education field - Create multiple donut chart to show the Attrition Rate by Gender for different Age group. - Build a dashboard. Video Lectures:	2	CIE+SEE+QUIZ	CO1,2 ,3,4



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6	IPL Dataset: - Create a calculated field to display match winner for each season in KPI. - Find out orange cap holder by selecting the appropriate filter & display in KPI. - Find out purple cap holder by selecting the appropriate filter & display in KPI. - Find out the count of tournament sixes by selecting the appropriate filter & display in KPI. - Find out the count of tournament fours by selecting the appropriate filter & display in KPI. - Create a horizontal bar chart to display matches won based on toss decision. - Create a donut chart to display the winning percentage based on toss decision. - Create a duplicate sheet for the team winners, orange cap & purple cap to display the respective images by uploading the images in Tableau repository. - Use features of horizontal containers, blank sheet & build an interactive dashboard.	2	CIE+SEE+QUIZ	CO1,2 ,3,4
7	Sales Profit Dataset: - Create KPI for sales profit and quantity using respective spark line. - Visualize sales & profit distribution by state (connecting with hexmap dataset). - Visualize average sales & profit by state along with respective state count, - Create a dynamic chart visualizing monthly states by segment. - Find out total sales by location & manager. - Build an interactive dashboard using the analysis.	2	CIE+SEE+QUIZ	CO1,2 ,3,4
8	Car Sales Dataset: - Create a bar chart on Dataset. - Create a line chart - Create a tree map - Build an interactive dashboard which includes all the information & charts.	2	CIE+SEE+QUIZ	CO1,2 ,3,4
9	Credit Card Complaint Dataset: - Create the following charts to track credit card complaints: - Total complaints along with rolling 12 months complaints with spark line chart. - Timely response with closed percentage progress bar. “In progress” along with its percentage – in progress spark line. - Create a dynamic chart to reflect trends as per the time duration (yearly, monthly, quarterly, weekly & daily). - Create a geographical density map with appropriate state & zip or pin code. - Create a filled map for the same with the colour gradient differentiation. - Sort top 10 issues based on its types. - Show data to display the company responses. - Create a calendar to show daily complaints. Use colour gradient to display. - Build an interactive dashboard which includes all the information	2	CIE+SEE+QUIZ	CO1,2 ,3,4



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	& charts.			
10	Visualizing ML Models using given Datasets	2	CIE+SEE+QUIZ	CO1,2 ,3,4

System Requirements:

1. System requirements are listed here under Tableau Desktop and Tableau Prep:

<https://www.tableau.com/products/techspecs>

2. The latest version of Tableau Desktop as well as Tableau Prep should be downloaded and installed from here: <https://www.tableau.com/tft/activation>

Text Books	
1	"Beautiful Visualization, Looking at Data Through the Eyes of Experts by Julie Steele, Noah Iliinsky"
2	"Information Dashboard Design: Displaying Data for At-a-glance Monitoring" by Stephen Few
3	Storytelling With Data: A Data Visualization Guide for Business Professionals" by Cole Nussbaumer Knaflic



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JAVALABORATORY

Course Code	21AI471	Credits:	01
L:T:P:S	0:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	AEC	Exam Hours	03

Course Objectives	
1	Understand the syntax, semantics, features and applications of Java Programming Language.
2	Learn the core concepts and construct using Java.
3	Understand the advanced concept and creating Java application.
4	Learn the method of creating connection between Java and Database.
5	Design and build simple Java applications.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Apply the concepts of Java to implement suitable solutions to real world problem.	L3
CO2	Analyze the given problem to provide a Java based solution.	L4
CO3	Design GUI application using Java Data Base Connectivity.	L4
CO4	Write clear and concise experiment reports that detail the methods, results, and conclusions of Java experiments conducted.	L5

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	-	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO4	-	-	-	2	-	-	-	-	-	-	-	-	-	-	2

Module	Topic	Hours	Assessment	CO
1	Generate the following abnormal conditions in main(): a) NumberFormatException b) OutOfStackMemoryError (invoke main() recursively) -> can you catch this as well? c) OutOfMemoryError - heap memory -> populate heap with 1 million reachable objects. Enable a catch block for each and see whether you are able to catch the exceptions. Print the stack trace and verify.	2	CIE+SEE+QUIZ	CO1
2	Implement the this() method for cube class containing 3 constructors.	2	CIE+SEE+QUIZ	CO1 CO4



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3	Develop a Java package to show dynamic polymorphism and interfaces.	2	CIE+SEE+QUIZ	CO2 CO4
4	Implement a method overriding and overloading using Java Inheritance.	2	CIE+SEE+QUIZ	CO2 CO4
5	Write a java program that implements a multi-thread application that has three threads. First thread generates random integer every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number.	2	CIE+SEE+QUIZ	CO1 CO4
6	Implement the event handling using Java AWT.	2	CIE+SEE+QUIZ	CO2 CO4
7	Develop a GUI application using Java swings.	2	CIE+SEE+QUIZ	CO3 CO4
8	Build a GUI Java program that connects to a database using JDBC and perform Add, Delete, Display and Exit operation	2	CIE+SEE+QUIZ	CO3 CO4
9	Create a servlet that uses Cookies to store the number of times a user has visited your servlet.	2	CIE+SEE+QUIZ	CO3 CO4

Text Books	
1	Java_-The Complete Reference, 11th edition, Herbert Schildt, Mc Graw Hill Education, 2019.
2	Programming with Java, E. Balagurusamy.
3	Head First Java: A Brain-Friendly Guide, Third Edition (Grayscale Indian Edition)

Reference Books	
1	J2SE Core Java, A.R.Kishore Kumar.



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UNIVERSAL HUMAN VALUES

Course Code	22UH48	Credits:	01
L:T:P:S	1:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	UHV	Exam Hours	03

Course Objectives	
1	Recognize the need for Value education harmony in self, family and society.
2	Understand the underlying the propositions of harmony in nature.
3	Verify the propositions and realize the implications of holistic understanding and professional ethics.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Recognize the importance of Value education.	L2
CO2	Understand the concepts of harmony in self.	L2
CO3	Develop skills for living in harmony and apply in professional life.	L3
CO4	Outline the principles of human values and examine its role in education	L2
CO5	Adapt to natural acceptance of human values build holistic understanding	L2
CO6	Test and verify the propositions on human values by self-exploration	L4

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	-	3	-	-	-	-	-	-	-	-	-	-	-	-	-

Module	Topic	Hours	Assessment	CO
1	Introduction to Value Education (3 hours) : Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education), Understanding Value Education, Self-exploration as the Process for Value Education, Continuous Happiness and Prosperity – the Basic Human Aspirations, Happiness and Prosperity – Current Scenario, Case studies, videos and real world examples	3	CIE+SEE+QUIZ	CO1 CO2
2	Harmony in the Human Being (3 hours) : Understanding Human being as the Co-existence of the Self and the Body, Distinguishing between the Needs of the Self and the Body, The Body as an Instrument of the Self, Understanding Harmony in the Self, Harmony of the Self with the Body, Case studies, videos and real world examples.	3	CIE+SEE+QUIZ	CO1 CO2



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3	Harmony in the Family and Society (3 hours) : Harmony in the Family – the Basic Unit of Human Interaction, 'Trust' – the Foundational Value in Relationship, 'Respect' – as the Right Evaluation, Other Feelings, Justice in Human-to-Human Relationship, Understanding Harmony in the Society, Vision for the Universal Human Order. Case studies, videos and real world examples.	3	CIE+SEE+QUIZ	CO3 CO4
4	Harmony in the Nature/Existence (3 hours): Understanding Harmony in the Nature, Interconnectedness, self-regulation and Mutual Fulfillment among the Four Orders of Nature, Realizing Existence as Co-existence at All Levels, The Holistic Perception of Harmony in Existence. Case studies, videos and real world examples.	3	CIE+SEE+QUIZ	CO3 CO4
5	Implications of the Holistic Understanding: a Look at Professional Ethics (3 hours) Natural Acceptance of Human Values, Definitiveness of (Ethical) Human Conduct, A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order, Competence in Professional Ethics Holistic Technologies, Production Systems and Management Models-Typical Case Studies, Strategies for Transition towards Value-based Life and Profession. Case studies, videos and real world examples. Presentation/Videos	3	CIE+SEE+QUIZ	CO5 CO6

Text Books	
1	A foundation course in HUMAN VALUES and Professional ethics; presenting a universal approach to value education- through self-exploration by, R R Gaur, R Sangal & GP Bagaria, Excel books Pvt. Ltd.
2	Professional Ethics & Human Values: Prof. D.R. Kiran, TATA Mc Graw Hill Education.

Reference Books	
1	Human Values: A. N. Tripathy (2003, New Age International Publishers)
2	Ethics in Engineering Mike W. Martin, Department of Philosophy, Chapman University and Roland Schinzinger, School of Engineering, University of California, Irvine.
3	Fundamentals of Ethics, Edmond G. Seebauer & Robert L. Barry, Oxford University Press